

Experiment 2

Title

Write a Program to Solve the 8-Queen Problem

Aim

To write a Python program to solve the 8-Queen problem using the Backtracking algorithm.

Objective

To understand the Backtracking technique and place eight queens on a chessboard such that no two queens attack each other.

Algorithm

Step 1: Create an empty 8×8 chessboard.

Step 2: Start placing queens from the first column.

Step 3: Check whether the current position is safe.

Step 4: If safe, place the queen.

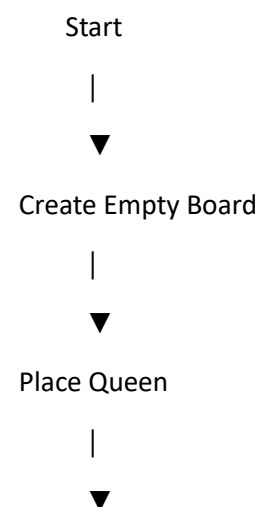
Step 5: Move to the next column.

Step 6: If no safe position exists, remove the previously placed queen (backtrack).

Step 7: Repeat until all eight queens are placed.

Step 8: Display the chessboard.

Flowchart



Position Safe?

| |

No Yes

| |

Try Next Row ▼

Place Next Queen

|

▼

All Queens Placed?

| |

No Yes

| |

Backtrack Display Board

|

▼

Stop

Python Program

N = 8

```
board = [[0] * N for _ in range(N)]
```

```
def safe(row, col):
```

```
    for i in range(col):
```

```
        if board[row][i]:
```

```
            return False
```

```
i, j = row, col
```

```
while i >= 0 and j >= 0:
```

```
    if board[i][j]:
```

```
        return False
```

```
i -= 1
```

```
j -= 1
```

```
i, j = row, col
```

```
while i < N and j >= 0:
```

```
    if board[i][j]:
```

```
        return False
```

```
    i += 1
```

```
    j -= 1
```

```
return True
```

```
def solve(col):
```

```
    if col == N:
```

```
        return True
```

```
    for row in range(N):
```

```
        if safe(row, col):
```

```
            board[row][col] = 1
```

```
            if solve(col + 1):
```

```
                return True
```

```
            board[row][col] = 0
```

```
    return False
```

```
if solve(0):
```

```
    for row in board:
```

```
        print(row)
```

```
else:
```

```
    print("No Solution")
```

Sample Output

[1, 0, 0, 0, 0, 0, 0, 0]

[0, 0, 0, 0, 0, 0, 1, 0]

[0, 0, 0, 0, 1, 0, 0, 0]

[0, 0, 0, 0, 0, 0, 0, 1]

[0, 1, 0, 0, 0, 0, 0, 0]

[0, 0, 0, 1, 0, 0, 0, 0]

[0, 0, 0, 0, 0, 1, 0, 0]

[0, 0, 1, 0, 0, 0, 0, 0]

Result

Thus, the 8-Queen problem was solved successfully using the Backtracking algorithm.

Conclusion

The Backtracking algorithm efficiently finds a valid arrangement of eight queens by checking safe positions and undoing incorrect placements whenever necessary.